Jacob Xing

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Education

Texas A&M University Aug. 2026 - May 2031
PhD in Electrical Engineering, Advisor: Dr. Sam Palermo, GPA: 4.0/4.0 *College Station, TX*
Honors: Graduate Fellow

University of Minnesota - Twin Cities Jan. 2025 - May. 2026
B.S. Electrical Engineering, GPA: 3.89/4.0 *Minneapolis, MN*
Honors: Summa Cum Laude, Dean's List, Distinction

Relevant Graduate Courses: Analog Integrated Circuit Design; RF/Microwave Engineering; VLSI Design; Semiconductor Devices; Digital Signal Processing; Microelectronic Fabrication

Professional Industry Experience

IBM May 2026 - Aug. 2026
Analog/Mixed-Signal IC Design Engineer Intern *Rochester, MN*

- Built MATLAB and Python code to characterize a 38.4 Gb/s PAM2 transmitter (Samsung SF2), enabling accurate S-parameter–based channel modeling for high-speed serial CDR simulations with microstrip lines, FFE/DFE, and eye diagram validation.
- Developed Verilog-A models of critical PLL building blocks (charge pump, loop filter, VCO) to accelerate top-level PLL simulation and design trade-off analysis.

Trusted Semiconductor Solutions May 2025 - Aug. 2025
Analog/Mixed-Signal IC Design Engineer Intern *Brooklyn Park, MN*

- Designed and successfully taped out a bandgap voltage reference core and startup circuit in Intel 18A using Synopsys tools, achieving a 37 ppm/°C TC across -55 to 125 °C ensuring robust temperature and process variation performances with 5 bit trimming capabilities.
- Led bring-up and configuration of Synopsys EDA environments, including licensing and environment variables.
- Designed a low-voltage differential signaling (LVDS) output buffer at 400 Mbps with 6.3mW power comsumption in GlobalFoundries 180nm and 45nm SOI technologies using Cadence Virtuoso.

Research & Academic Projects

Dr. Palermo's Research Group Aug. 2026 - Present
Analog/Mixed-Signal Research Assistant *College Station, TX*

- Currently conducting research on high-speed eletrical and optical interconnects

Dr Harjani's Research Group Aug. 2025 - May 2026
Analog IC Undergraduate Research Assistant *Minneapolis, MN*

- Architected an analog readout scheme for a 60 FPS, 24 megapixel CMOS image sensor by designing a CDS circuit and single-slope ADC in LTSPICE and Cadence Virtuoso.
- Emulation of CMOS image sensor process photodiodes with TSMC 16nm PDK to ensure accurate simulations.

Biquad Switched-Capacitor Active Filter | *Cadence Virtuoso, RFIC* Aug. 2025 - Present

- Specced and designed a biquad LPF for Wi-Fi receivers with programmable BW from 1.6GHz, sub $10nV/\sqrt{Hz}$ input referred noise, Op-Amp gain of 80 dB and sub 1mW power consumption in FinFET TSMC 16nm.

Reconfigurable Systolic Array for Edge AI | *Verilog, PrimeTime PX* Jan. 2026 - May 2026

- Architected and designed a reconfigurable systolic array for matrix multiplication and AI inference, achieving a 50% improvement in silicon efficiency over the baseline design in GlobalFoundries 22nm.

Selected Publications

- C. Kim, P. Kreye, A. Efe, R. Yin, C. Li, A. Kumar, T. Islam, Z. Zeng, N. Prova, A. Vanasse, X. Li, **J. Xing**, et al, "COBIFIVE: A Five-Core Physics-based Ising Computing Chip", Nature Electronics, 2025 (Under Review).

Technical Skills

Cadence Virtuoso, Cadence Spectre, Synopsys, Primewave, Primesim, Verilog-AMS, HSPICE, Altium, Intel Quartus Prime, ModelSIM, Vivado, Matlab, LTSPICE, VS Code, ADS Keysight, Microsoft Excel, PowerPoint